Journal of Dentistry 144 (2024) 104970

Contents lists available at ScienceDirect

Journal of Dentistry

journal homepage: www.elsevier.com/locate/jdent

Performance comparison of multifarious deep networks on caries detection with tooth X-ray images

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ARTICLE INFO

Keywords:
Caries diagnosis
Caries detection
Radiographically diagnosis
Dental radiographs
Deep learning

ABSTRACT

Objectives: Deep networks have been preliminarily studied in caries diagnosis based on clinical X-ray images. However, the performance of different deep networks on caries detection is still unclear. This study aims to comprehensively compare the caries detection performances of recent multifarious deep networks with clinical dentist level as a bridge.

Methods: Based on the self-collected periapical radiograph dataset in clinic, four most popular deep networks in two types, namely YOLOv5 and DETR object detection networks, and UNet and Trans-UNet segmentation networks, were included in the comparison study. Five dentists carried out the caries detection on the same testing dataset for reference. Key tooth-level metrics, including precision, sensitivity, specificity, F1-score and Youden index, were obtained, based on which statistical analysis was conducted.

Results: The F1-score order of deep networks is YOLOv5 (0.87), Trans-UNet (0.86), DETR (0.82) and UNet (0.80) in caries detection. A same ranking order is found using the Youden index combining sensitivity and specificity, which are 0.76, 0.73, 0.69 and 0.64 respectively. A moderate level of concordance was observed between all networks and the gold standard. No significant difference ($p > 0.05$) was found between deep networks and between the well-trained network and dentists in caries detection.

Conclusions: Among investigated deep networks, YOLOv5 is recommended to be priority for caries detection in terms of its high metrics. The well-trained deep network could be used as a good assistance for dentists to detect and diagnose caries.

Clinical Significance: The well-trained deep network shows a promising potential clinical application prospect. It can provide valuable support to healthcare professionals in facilitating detection and diagnosis of dental caries.

1. Introduction

Dental caries is a prevalent global health issue, affecting approximately 2.3 billion individuals across different severity levels [1]. The standard treatment protocol for dental caries primarily involves the excision of decayed tissue within the cavity, followed by restoration using materials such as composite resin. Consequently, determining whether a carious lesion has formed and establishing the endpoint for caries removal remains one of dentists' crucial clinical skills [2]. Caries can only be identified and diagnosed after its formation. According to the diagnostic criteria from WHO, dental caries can be diagnosed when there are defects present in the pit, groove, or smooth surface of the tooth, as well as softening observed on the floor or wall of the tooth surface [3]. Current caries detection methods encompass clinical inspection, probing, and imaging examinations such as periapical radiography and occlusal wing radiograph [4,5]. It was reported there was no significant difference between experienced dentists and students in caries diagnosis by means of bitewing radiographs [4], which also indicated that well designed and trained artificial intelligence (AI) would be helpful and welcome for dentists. Additionally, some non-invasive techniques such as optical coherence tomography (OCT) also arise for early caries detection, though the scope of their use is still limited [6].

Among these methods, the combination of imaging and intraoral

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https://doi.org/10.1016/j.jdent.2024.104970
Received 23 September 2023; Received in revised form 11 March 2024; Accepted 24 March 2024
Available online 30 March 2024
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examination remains a conventional and widespread clinical approach for detecting and diagnosing dental caries. Proximal caries, particularly in their early stages, pose challenges for direct diagnosis in the clinic due to their concealed location and lack of apparent defects or clinical symptoms. Consequently, reliance on density changes observed on X-ray films becomes crucial [5]. Studies have indicated that periapical radiographs and bitewing radiographs exhibit greater sensitivity than clinical examination in identifying proximal caries [7,8]. However, it has been reported that diagnostic accuracy is operator-dependent [9] and often subjectively influenced with unstable judgment of results [10]. As mentioned before, experienced dentists will not perform significantly better than final-year dental students in caries diagnosis. [4] Moreover, manual inspection suffers from low efficiency as a drawback. All these raise the need of assistant caries diagnosis with machines and AI.

With the swift advancement of AI technology, the feasibility of devising an automated caries detection methodology characterized by heightened efficiency and precision has emerged, thereby augmenting clinical caries diagnosis. Initially, researchers employed classical unsupervised learning model [11] or shallow feed-forward artificial neural network [12] to ascertain the presence of caries, utilizing panoramic or periapical dental X-ray imagery. These approaches necessitated either traditional image processing techniques, such as edge detection, which lack adaptability to diverse image conditions, or manual feature selection, a process that is both inefficient and challenging in maintaining comprehensive feature integrity.

Deep networks, distinguished by their powerful self-extraction capability of feature representation, exhibit more robustness against complex interferences compared to traditional AI methods [13]. Over the past decade, deep convolutional neural network (CNN) have achieved significant breakthroughs in various medical imaging tasks [14,15], including the field of dentistry [16,17]. Recent dental researches have started to use deep networks to detect caries [18]. Predominantly, three types of deep networks are utilized in this domain: classification networks [19], object detection networks [20–22] and semantic segmentation networks [13,21,23–27]. Lee et al. [19] used the GoogLeNet deep classification CNN for diagnosing dental caries in periapical radiographs. However, this classification network was limited to identifying the presence of caries in images, without the capability to detect and locate decay, thus constraining its clinical utility. With the emergence of the prominent segmentation network UNet [28], numerous researchers have explored its variants to detect and segment caries. For instance, Cantu et al. [25] applied the original UNet on bitewings and revealed a sensitivity of 0.75 that was significantly higher than dentists. Similarly, Lee et al. [24] implemented UNet for early caries detection in bitewing radiographs and found that with the help of the network clinicians could improve the detection sensitivity. Casalegno et al. [26] presented a variant of UNet with VGG16 for the automated detection and localization of dental lesions in near-infrared transillumination images. Recently, Zhu et al. [23] constructed a U-shape network with the additional full-scale axial attention module to segment caries in oral panoramic radiographs. Ying et al. [13] proposed an innovative U-shaped network for caries segmentation, incorporating vision Transformer, dilated convolution, and feature pyramid fusion methods, claiming a sensitivity of 0.94 in caries-level detection in their study. Besides segmentation networks, there are other researches focusing on object detection networks for caries detection. Bayrakdar et al. [21] investigated caries lesions detection performance with a VGG-16-based object detection network on dental bitewing radiographs, achieving a sensitivity of 0.84. In a related study, Bayraktar et al. [20] revealed a sensitivity of 72.26 % in diagnosing interproximal caries lesions on digital bitewing radiographs by modifying the popular YOLO (You Only Look Once) object detection network. Additionally, Zhang et al. [22] screened dental caries from oral photographs by adapting the SSD (Single Shot MultiBox Detector) object detection network. Despite the aforementioned studies, it is still a challenge for clinical dentists in understanding and selecting the appropriate deep network for specific caries detection applications. This difficulty arises from the scarcity of studies that report on the performance disparities among different network types in caries detection. Due to inconsistencies in training datasets, hyperparameters and so on, the performance metrics, such as sensitivity, cannot be compared directly across different researches. Moreover, it is also unclear whether there will be a significant difference of the caries diagnosis between deep networks and clinical dentists.

To assist clinicians and researchers in selecting appropriate deep networks for caries detection, particularly for proximal caries or multi-faceted caries with proximal lesions, we conducted a thorough comparative analysis of various deep network architectures based on the same X-ray image dataset. The main aim of this study is to reveal performance differences between multiple type deep networks and between networks and dentists for caries detection in radiographs. The famous segmentation network UNet [28] and its improved variant Trans-UNet with Transformer [29], and two popular deep object detection networks, namely YOLOv5 [30,31] and DETR (Detection Transformer) [32], were evaluated based on key metrics including precision, sensitivity, specificity, F1-score and Youden index. Furthermore, five dentists performed the caries detection on the same test dataset. Their results will further support the result in the literature [4,33] that there was no significant difference between experienced and inexperienced dentists in caries detection with radiographs, and are then presented as a comparative benchmark to enhance understanding of the deep networks' performance. Keeping the aim of this study in mind, we put forward the null hypothesis (H_0) that there was no significant difference between multiple deep networks and between networks and dentists in caries detection with radiographs. Statistics based on caries detection results were conducted to reject or not reject the null hypothesis.

2. Method

2.1. Tooth X-ray image dataset

This study was approved by the bioethics committee of our hospital (No. 2021–58R). The main task was to identify the presence and location of the caries in periapical radiographs. All tooth X-ray images were obtained from adult patients with yellow race who had signed an informed consent form during the years from 2021 to 2023. These dental radiographs were generated by a radiographic machine (FOCUS; Instrumentarium Dental PaloDEx Group Oy, Tuusula, Finland). The inclusion criteria were: teeth with obvious caries on the adjacent surface, the depth of caries lesions was medium caries and above, no developmental abnormalities, no malocclusion, and no periodontal diseases. All caries were further confirmed with clinical examination by senior stomatologists. While the exclusion criteria were: patients with severe periodontal diseases, irregular dentition, or dentition loss; patients with mental diseases, critical diseases, pregnant women, and other uncooperative patients. This study did not involve vulnerable groups. At last, total 392 periapical radiographs were collected as the original images of this study. Afterwards, these images were randomly divided into two subsets: 346 images as the training and validation dataset, and the remaining 46 images as the testing dataset.

2.2. Testing sample size estimation

To verify whether the sample size of the test dataset was sufficient, the sample size estimation was carried out for this single sample diagnostic test [34].

$$n = p(1-p)\left(\frac{Z_{1-\alpha/2}}{\delta}\right)^2 \tag{1}$$

Where $Z_{1-\alpha/2} = 1.96$ was determined by the look-up table with a significance level of α=0.05; δ was the allowable error which is 10 % in this study; p was the expected sensitivity. A higher value of p would yield a smaller sample size, and therefore to guarantee an adequate sample size a low value of p should be selected. Through reviewing related references of deep networks for caries detection [19,25], and simultaneously combining the preliminary experiments of our research, the sensitivity was found to be no less than 0.70. To further leave sufficient margin, p was finally assumed to be 0.60 here. According to above, the required sample size was determined to be at least 93 teeth. Given that there were 135 teeth in the 46 radiographs of the testing dataset, the sample size was deemed sufficient for the study's purposes.

2.3. Labeling

A group of three senior stomatologists was in charge of labeling the ground truth for all the X-ray images. In the first stage, two professional stomatologists with more than 10-year clinical experience fully discussed and then meticulously delineated the caries decay mask together. Subsequently, a more senior stomatologist with over 20-year clinical experience reviewed and revising these masks, which were then regarded as the final ground truth (gold standard) labels. For object detection networks, the ground truth was transformed to a bounding rectangle from the corresponding mask. Furthermore, five additional dentists from the same clinical department, with 7-year (Dentist A), 5-year (Dentist B), 3-year (Dentist C), 1-year (Dentist D) and less than 1-year (Dentist E) clinical experience respectively, independently labeled the testing dataset for the following comparative analysis. These dentists had been trained in reading periapical radiographs in the radiology department and had developed clinical diagnosis and treatment expertise through daily practice. Notably, no specialized intensive training in caries detection was carried out for the dentists in this study. All participating experts were trained to use the annotation tool (LabelMe; MIT CSAIL, Cambridge, MA, USA) [35] before the labeling task.

2.4. Training data augmentation

To enhance sample diversity and improve model generalization performance, data augmentation was performed on the training dataset. We applied the conventional offline augmentation operations, including translation, flip, mirror, rotation, vertical perspective deformation, oblique quadrangular perspective deformation, region warping, and shear mapping transformation, for the training images and their corresponding labels with the help of a library package (Augmentor; Medical University of Graz, Graz, Austria) [36]. Afterwards, all augmented images were validated by the labeling stomatologists and meaningless

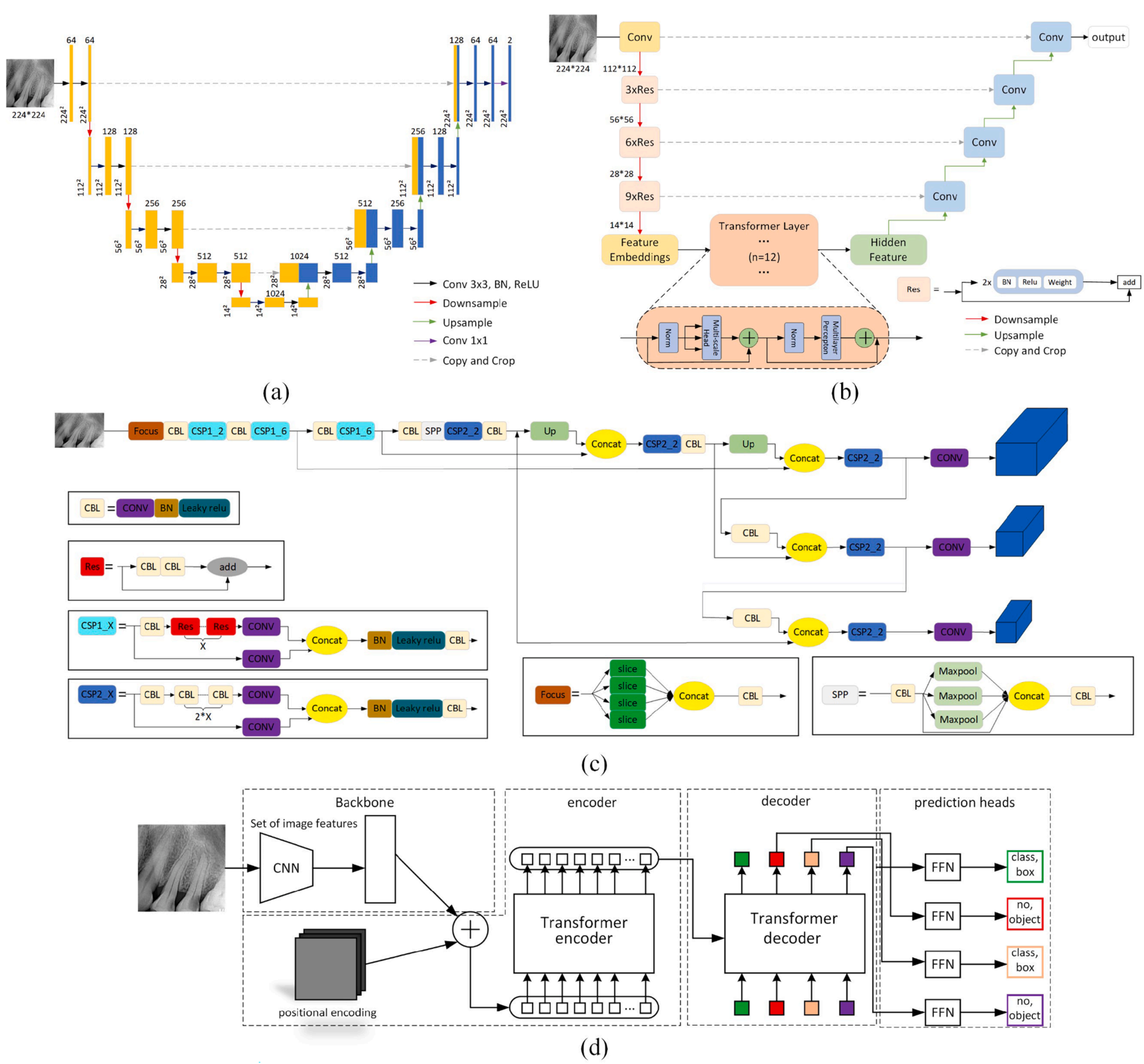

Fig. 1. Deep network architectures used in this comparison study. (a) UNet; (b) Trans-UNet; (c) YOLOv5; (d) DETR.

images were removed. At last, a total number of 1730 images were formed to be the training dataset, among which 10 % were used as validation during training sessions.

2.5. Deep network architectures

For the comparative analysis, four deep networks were selected from recent typical segmentation and object detection networks, which included: UNet [28], Trans-UNet [29], YOLOv5 [30,31] and DETR [32]. The first two networks were classical segmentation networks, whereas the latter two were widely used object detection networks. Their network architectures are depicted in Fig. 1.

As shown in Fig. 1(a), UNet was a U-shaped deep convolutional network composed of the encoder that extracted feature maps and the decoder that restored the image size with upsampling. Skip connections were introduced between the encoder and decoder to fuse deep and shallow layer features. Trans-UNet, illustrated in Fig. 1(b), was a variant of UNet and had similar structure, with mainly two aspects of improvements. The first improvement was to adopt the 50-layer ResNet instead of previous shallow convolution network as the backbone of the encoder for better feature extraction. And the second improvement was to bring in the visual Transformer module after the backbone to consolidate the linkage of contextual information.

YOLOv5, a mainstream one-stage target detection network, was renowned for its rapid processing capabilities while maintaining commendable accuracy. Its architecture mainly composed of backbone, neck and head along with their functional structure. As shown in Fig. 1 (c), the backbone was responsible for feature extraction, and after that the neck utilized and fused the features, and the head output the detection result in the end. There were four release versions of YOLOv5 according to network complexity, and in this study the YOLOv5s was used. The last network DETR was a typical Transformers based object detection network. As depicted in Fig. 1(d), its prominent structure was the Transformer encoder and decoder. Other main components contained a CNN backbone for feature extraction and feed forward networks (FFN) to generate the final object detection prediction.

2.6. Evaluation metrics

In this study, a comprehensive set of performance evaluation metrics was employed, including precision, sensitivity, specificity, F1-score and Youden index. We adopted the tooth-level statistical method from the literature [25]. Concretely, the caries detection result within an individual tooth was classified as true positive (TP), false positive (FP), false negative (FN) and true negative (TN), as shown in Fig. 2. To further clarify, it was regarded as TP if there was an overlap between the predicted area (a blob in segmentation or a box in object detection) and the corresponding ground truth, while it was regarded as TN if no prediction was made within a healthy tooth. Finally, TP, FP, FN and TN in one tooth were normalized and then summed over all teeth in the testing images to obtain the overall values of TP, FP, FN and TN.

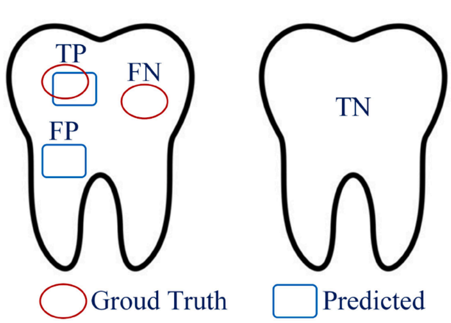

Fig. 2. Schematic diagram of true positive TP, false positive FP, false negative FN and true negative TN within a single tooth.

Based on the overall values of TP, FP, FN and TN, the precision, sensitivity, specificity, F1-score and Youden index [28] metrics were calculated using the following expressions. The last two metrics F1-score and Youden index were synthetical indicators that involved the first three metrics.

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{3}$$

$$Specificity = \frac{TN}{TN + FP} \tag{4}$$

$$F1 - score = \frac{2 \cdot Precision \cdot Sensitivity}{Precision + Sensitivity} \tag{5}$$

$$Youden\ index = sensitivity\ +\ specificity\ -\ 1 \tag{6}$$

Besides above metrics, the average dice similarity of all testing images was also provided for segmentation network and compared dentists. Dice similarity was a measure of the overlap and matching degree between the corresponding segmented region and the labeled ground truth mask [13], which was often used to evaluate the performance of the segmentation network. It is noted that the object detection network did not delineate caries area and thus had no dice similarity.

2.7. Statistics

Cohen's Kappa (κ) consistency statistic was used to reveal the agreement between the observer and the gold standard (ground truth) regarding the detection result of caries. Moreover, the McNemar Chi square test based on the statistical results of TP, FP, FN and TN was employed to assess the significance of differences for caries detection performances between observers. The paired *t*-test was adopted to evaluate the significance of differences between observers' average dice similarities in caries segmentation. All calculations were conducted in the SPSS 23.0 statistical software (Statistical Package for the Social Sciences; IBM, New York City, New York State, USA). And $p < 0.05$ was considered statistically significant.

2.8. Training implementation

The deep networks were implemented in the Pytorch 1.9.0 (Pytorch; Meta AI, New York City, New York State, USA) deep learning framework with Python (Python; Python Software Foundation, Beaverton, OR, USA) as the interface language. The hardware of this study was a computer workstation with an Intel i7–7820 processor (i7–7820; Intel, Santa Clara, CA, USA) and a NVIDIA Titan V graphics card (Titan V; NVIDIA, Santa Clara, CA, USA), and the software platform environment was Ubuntu18.04 (Ubuntu; Canonical Ltd., London, United Kingdom) + CUDA10.2 (CUDA; NVIDIA, Santa Clara, CA, USA) + cuDNN8.8.1 (cuDNN; NVIDIA, Santa Clara, CA, USA). During training, the input image size for the object detection networks was 400 × 400, while it was resized to 224 × 224 for the segmentation networks. The hyperparameters for every network were carefully selected to get an optimal training result. At last, the training configurations for all networks, including batch size, initial learning rate, optimizer and epochs, were determined, as detailed in Table 1. It is noted that all networks were pretrained, except for UNet.

3. Results

3.1. Metrics

After training, according to the evaluation metrics described in section 2.6, caries detection results (TP, FP, FN and TN) on tooth-level were

Table 1
Training setup for the networks.

	Segmentation networks		Object detection networks	
	UNet	Trans-UNet	YOLOv5	DETR
Image size	224 × 224	224 × 224	400 × 400	400 × 400
Batch size	12	12	12	12
Initial learning rate	0.005	0.005	0.01	0.0001
Learning rate schedule	Adaptive	Linear	Cosine annealing	Piecewise constant
Optimizer	RMSProp	SGD	Adam	AdamW
Epochs	200	200	300	1000

counted, based on which the metrics of precision, sensitivity, specificity, F1-score and Youden index were calculated. These metrics for the comparative dentists and networks are listed in Table 2, with the highest and lowest scores of each metric in column highlighted in bold and underlined respectively. It can be seen from the table that the object detection network DETR achieved the highest scores of 0.95 in precision and 0.96 in specificity, while Dentist A reached the highest sensitivity and F1-score that are 0.89 and 0.90 respectively. Although DETR had high scores in precision and specificity, it had the lowest sensitivity. Besides sensitivity, the lowest scores of other metrics occurred in Dentist E.

The Youden index was adopted to synthetically compare the diagnostic capabilities of the networks and clinicians. The calculated Youden index values for each participant are listed in the last column. As can be seen from the table, Dentist A achieved a highest Youden index value of 0.81, followed by the object detection network YOLOv5, which attained a value of 0.76. The Youden indices of Dentist B, the segmentation network Trans-UNet and Dentist C were 0.74, 0.73 and 0.72, respectively. DETR, which was outstanding in terms of precision and specificity, had a Youden index of only 0.69. The Youden index of the segmentation network UNet fell in between the indices of the Dentists D and E. As for networks, the performance order from highest to lowest was YOLOv5 > Trans-UNet > DETR > UNet. However, no significant difference ($p > 0.05$) was found between these networks' performances according to the McNemar tests. When comparing YOLOv5, which had the highest score among networks, with every dentist, there was no significant difference ($p > 0.05$) found. In addition, the difference between the detection results of the five dentists was also found to be insignificant ($p > 0.05$). In summary, although the metrics show differences regarding the caries detection results of networks and dentists, these differences were not statistically significant.

Table 4
Dice similarities of the dentists and the segmentation networks on the testing dataset.

	Dentists					Segmentation Networks	
	A	B	C	D	E	Trans-UNet	UNet
Dice	0.751	0.744	0.732	0.677	0.629	0.680	0.603

The results of Cohen's Kappa consistency statistical test between the observers and the gold standard is listed in Table 3. It is found that YOLOv5 showed the best kappa consistency among networks with a value of 0.763 ($p < 0.05$), which was between the value of the Dentist A (0.807, $p < 0.05$) and Dentist B (0.734, $p < 0.05$), followed by Trans-UNet (0.713, $p < 0.05$), DETR (0.687, $p < 0.05$), and UNet (0.641, $p < 0.05$). Nevertheless, all the network's caries detection results were in moderate strength agreement with the gold standard.

Table 4 lists the average dice similarity coefficients of segmentation networks and dentists relative to the ground truth masks. It can be seen from the table that the dice similarity of the Trans-UNet was 0.680, which was between the dice similarities 0.732 and 0.677 of the Dentist C and D respectively. The dice similarity of the Dentist A achieved a highest value of 0.751, followed by that of the Dentist B which was 0.744. While the UNet had a lowest dice similarity, which is only 0.603. Comparing dentists and networks, significant differences ($p < 0.05$) of the dice coefficients were found in the following groups according to the paired *t*-test results: (Dentist A vs. Trans-UNet), (Dentist A vs. UNet), (Dentist B vs. UNet), and (Dentist C vs. UNet). In addition, the difference between Trans-UNet and UNet was also statistically significant ($p < 0.05$).

3.2. Caries detection examples

Fig. 3 presents selected correct examples of the caries detection result by all dentists and networks from the testing X-ray image dataset. The ground truth masks are shown in red in the first row, while the output masks and detection box of the comparison dentists and networks are displayed in magenta in the rest rows, with one row for one dentist or network. In these examples, all caries locations were targeted correctly by whether dentists or networks. However, the outlines of decayed areas were diversified for both dentists and segmentation networks, none of which fit perfectly with the ground truth mask. Compared to those outlined by dentists, the segmented boundary of Trans-UNet and UNet seemed to have a greater difference from the ground truth. It is noted that there are no segmented contours but only predicted boxes of caries for the object detection networks YOLOv5 and DETR. Despite the irregular distribution of caries, the figure exhibits that the networks

Table 2
Performance comparison of the dentists and the networks on the testing dataset.

		Precision	Sensitivity	Specificity	F1-score	Youden Index
Dentists	A	0.91	**0.89**	0.91	**0.90**	**0.81**
	B	0.92	0.81	0.94	0.86	0.74
	C	0.92	0.78	0.93	0.85	0.72
	D	0.80	0.83	<u>0.82</u>	0.82	0.65
	E	<u>0.79</u>	0.79	<u>0.82</u>	<u>0.79</u>	<u>0.61</u>
Segmentation Networks	Trans-UNet	0.90	0.81	0.92	0.86	0.73
	UNet	0.85	0.76	0.88	0.80	0.64
Object Detection Networks	YOLOv5	0.93	0.82	0.94	0.87	0.76
	DETR	**0.95**	<u>0.72</u>	**0.96**	0.82	0.69

Table 3
Results of Cohen's Kappa consistency statistical test ($p < 0.05$).

	Dentists vs. Ground Truth					Deep Networks vs. Ground Truth			
	A	B	C	D	E	YOLOv5	Trans-UNet	DETR	UNet
Kappa (κ)	0.807	0.734	0.705	0.646	0.612	0.763	0.713	0.687	0.641

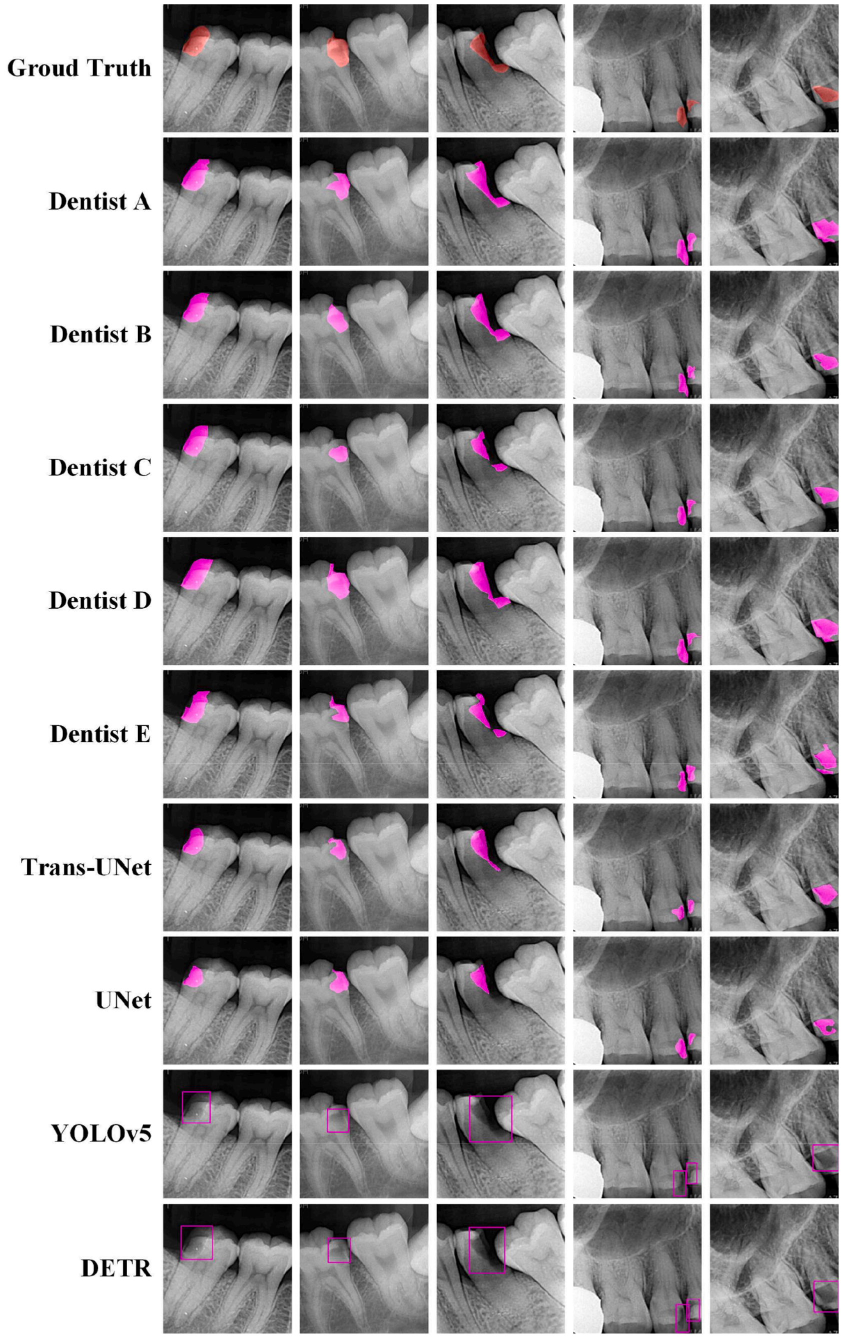

Fig. 3. Correct examples of the caries detection results for all the dentists and networks from the testing dataset.

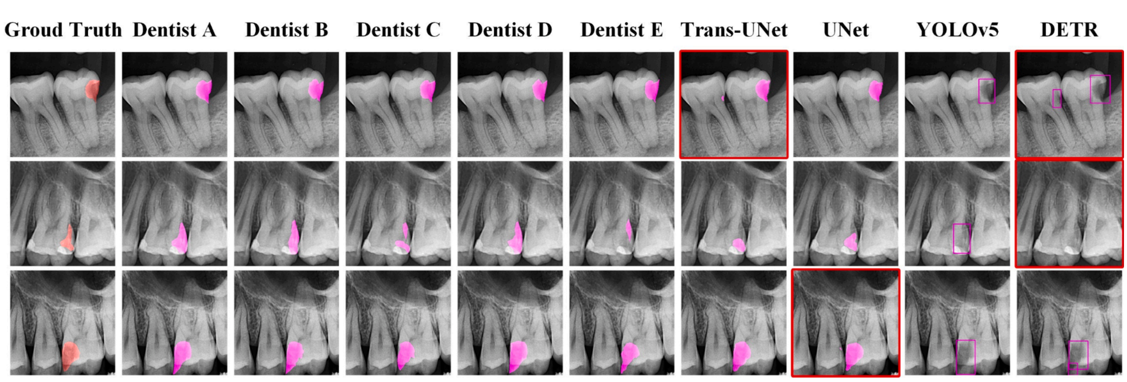

Fig. 4. Examples which the networks make mistakes in detecting the caries while all the dentists are correct. The incorrect examples are highlighted with red boxes.

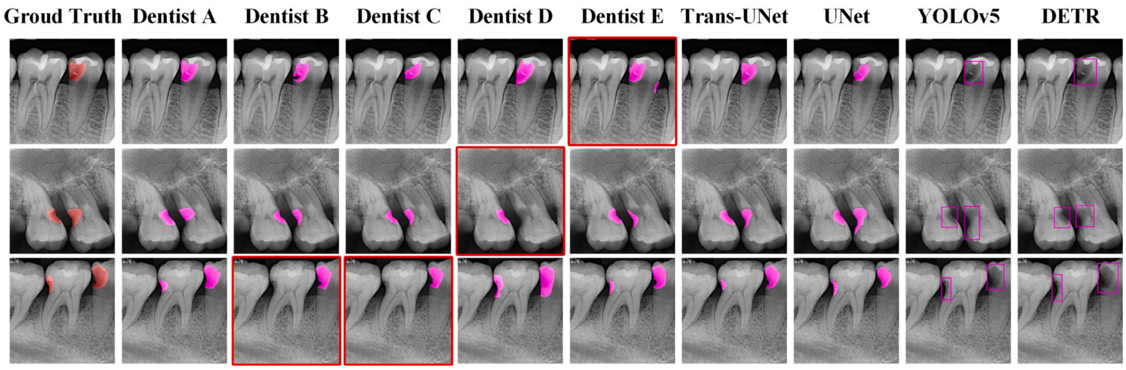

Fig. 5. Examples which the dentists make mistakes in detecting the caries while all the networks are correct. The incorrect examples are highlighted with red boxes.

behaved well in caries detection, aligning with the results in Table 2.

Fig. 4 and 5 show several examples where either the networks or the dentists have inaccurately identified caries. The detection results for each participant are displayed in columns, with the first column showing the ground truth. The incorrect examples are highlighted with red boxes. As shown in Fig. 4, Trans-UNet, UNet and DETR all predicted redundant caries (FP). Trans-UNet and DETR made wrong prediction at the same position, while UNet also made similar wrong prediction. Notably, the mis-predicted caries areas were relatively small, as can be seen from the output masks made by Trans-UNet and UNet. There was also a missed caries detection (FN) of DETR in the middle row of the last column in Fig. 4. It's worth mentioning that these mistakes were all avoided in YOLOv5.

There were also some cases that the networks performed better than the dentists. As shown in Fig. 5, the redundant caries prediction (FP) also occurred in Dentist E. Additionally, Dentist B, C and D all missed to detect a part of caries. Whereas the networks performed quite well in these radiographs as they predicted all the caries correctly.

4. Discussion

The null hypothesis (H0) that there was no significant difference between multiple deep networks and between networks and dentists in caries detection with radiographs was not rejected by the statistic test results ($p > 0.05$). In other words, the caries detection performance of dentists would not be significantly better than deep networks. All the networks' caries detection performances were in moderate strength agreement with the gold standard. This means a well-trained deep network could be a good assistance in the caries diagnosis for dentists.

In the context of rapid advancements in deep learning, an increasing number of dental researchers are applying this technology to caries detection. However, a general lack of deep network expertise among dentists poses challenges in selecting the most appropriate model for specific application. On the other hand, many dentists remain uncertain about the current capabilities of deep networks in caries detection, particularly in comparison to the diagnostic skills of clinically experienced dentists. This study aims to methodically compare the performance of various deep network types—including segmentation networks UNet and Trans-UNet, and object detection networks YOLOv5 and DETR—with clinical dentists in caries detection, using our custom tooth X-ray image dataset. Through the tooth-level result statistics, we evaluated key metrics like sensitivity and specificity for both the networks and the dentists, employing the Youden index for a comprehensive performance assessment. Our results revealed that there was no statistically significant difference in caries detection between well-trained deep networks and dentists. This study offers a valuable reference for dentists and researchers in understanding and selecting suitable deep networks for caries detection applications.

The five dentists for comparison in this study were selected with different years of clinical experience. And metrics such as Youden index showed caries detection difference between them. However, this difference was turned out to be not statistically significant. This finding was consistent with previous studies by Wrbas et al. [4,33], in which no significant difference (ANOVA) was found in the ability to diagnose caries from bitewing radiographs between the groups of dentists and final-year students, and the groups of students received and not received additional training. It indicates that if the basic training has sufficient quality, no greater improvements will be possible from dentists' daily caries detection with radiographs. Moreover, the specific experience of the involved dentist is not representative and his/her result in this study cannot be generalizable. As a result, the experiences of participating dentists are only for reference, and it is inappropriate to directly compare experiences.

Although statistics indicated there was no significant difference in caries detection between the investigated deep networks, there were numerical differences in terms of key metrics. The object detection network YOLOv5 showed the highest sensitivity, F1-score and Youden index among networks, and the second highest precision and specificity, which demonstrated its good overall performance. Notably, DETR exhibited the highest precision and specificity among all networks, but with the lowest sensitivity. Generally, the networks behaved well in precision and specificity, indicating that the caries predicted by networks is more likely to be actual, which from another perspective also implies a tendency of the networks to miss ambiguous caries. Among segmentation networks, all metrics of Trans-UNet were higher than those of UNet, showing the positive effect of network's structure improvement. In conclusion, YOLOv5 emerges as the most appropriate network choice for caries detection in this study. When segmentation of caries is also required, Trans-UNet is preferable. However, it is emphasized that a network must be used as assistance or help, and the final decision of caries in clinic should always be made by the dentist.

In specific instances, such as with relatively straightforward and evident caries like deep caries, all evaluated dentists and networks demonstrated proficient identification skills. This suggests that the trained networks possess a foundational level of performance. In addition, while most dentists missed some decays at the neck of teeth, the networks had been notably successful in identifying these decays. Accurately distinguishing between neck decay and natural depression in clinic is challenging, and misidentification or omission may occur in dentists. Although networks may also misdiagnose these dental decays, their heightened sensitivity in these typically challenging-to-observe regions offers valuable assistance to dentists, irrespective of their years of clinical experience.

Although from the perspective of dental caries detection the performance of well-trained YOLOv5 or Trans-UNet had no significant difference with dentists, these networks fell short in accurately delineating caries areas. A significant difference in the dice coefficient was found between Trans-UNet and Dentist A. This outcome suggests if the goal is to segment the caries area, the current segmentation network still has much room for improvement. It is noteworthy that the difference in dice similarity between Dentist A and Dentist D, or Dentist A and Dentist E was significant, indicating that dentists exhibit considerable subjectivity in determining the area of dental decay. Consequently, achieving further improvement in dice similarity will be challenging.

This comparative study offers an in-depth analysis of the application of deep networks in caries detection, providing valuable insights for researchers in selecting appropriate networks. The findings underscore substantial potential of deep networks for clinical application in this field. Deep networks, performing no significant difference with dentists in caries detection, can assist dentists in better and fast identifying caries in radiographs, and serve as an educational tool in medical colleges. The results indicate that experienced dentists may overlook certain subtle caries as well, which the networks can identify, thereby potentially reducing diagnostic omissions and increasing accuracy. Meanwhile, the speed of the network to identify caries is superior. For instance, the reasoning time of YOLOv5 for an X-ray film is only about 0.07 s (with a CPU of i7–8700k), which is about 50 times faster than the average several seconds taken by dentists. Consequently, the integration of these networks into dental practices can substantially decrease the time dentists spend analyzing radiographs, thereby improving their overall work efficiency.

Finally, it is pointed out that there are still several shortcomings and limitations in this study. The first one is the limitation of the image data set. Only periapical radiographs were used in data set, and other radiograph types such as bitewing were not included. This is due to periapical radiographs were much more commonly taken in our hospital, and the dataset for training was easier to access. Although using only periapical radiographs is reasonable for our current comparison study, to include other types such as bitewing radiographs could further make use of their characteristics and enrich the dataset. This will be considered in the future work. In addition, the size and diversity of the data set still need to be strengthened. More dental X-ray images obtained from different types of devices with various parameter conditions in different hospitals and regions to form a large-scale data set can help improve the diversity of caries samples and further verify the performance reliability and stability of the network. Second, as mentioned before, the performance and experience of the dentists included in this study is not representative and cannot be generalized. To make the comparative conclusion more general, more dentists from various organizations in various regions can be included in the comparison study and large-scale statistics should be conducted to eliminate the occasionality and individual bias. Third, only two typical networks from the object detection networks and the segmentation networks respectively were selected for comparison. With the rapid development of the current deep network technology, newer and more advanced networks can be used for caries detection, which may improve the performance of the network to a better level. At last, it is emphasized that currently the deep network cannot be considered an independent diagnostic method but rather serves as an adjunctive tool in caries diagnosis. It constitutes an image reading and understanding procedure in the radiography caries diagnosis method and can also be used in other caries diagnosis method such as OCT. The caries detection in this study mainly refers to identify if and where caries exists in the X-ray films. Therefore, the comparison between deep networks and doctors is only limited to this aspect. The caries detection result of deep networks on radiographs should be confirmed by clinical examination in actual clinical applications. As the caries detection in X-ray films relies on density changes, due to influence such as artifacts in X-ray films, it may also lead to false negatives diagnosis without any clinical examination. It is noted that the caries in this study is all ensured by the clinical investigation.

5. Conclusion

The statistical analysis shows no significant performance difference in caries detection with periapical radiographs between the investigated multifarious deep networks. However, YOLOv5 is recommended to be priority among these networks for caries detection in terms of its high metrics. Findings also indicate there is no significant difference in caries detection between the well-trained deep network and the dentists. Therefore, the deep network could be used as a good assistance for dentists to detect and diagnose caries in clinic. Our findings could be further generalized by investigations involving more future improved deep networks and dentists from different hospitals and institutions.

CRediT authorship contribution statement

Shunv Ying: Writing – original draft, Resources, Methodology, Funding acquisition, Data curation, Conceptualization. **Feng Huang:** Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Xiaoting Shen:** Resources, Data curation. **Wei Liu:** Writing – review & editing, Validation, Resources, Data curation. **Fuming He:** Writing – review & editing, Validation, Supervision, Resources.

Declaration of competing interest

The authors declared that they have no conflicts of interest. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Funding

This study was funded by General Scientific Research Project of Education Department of Zhejiang Province (Grant No. Y202147147).

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